HDOC DAY 49

PROGRAM

Write a C program that reads a positive integer n. Pass n to a function named pivot_integer() using its address. Use pointers to pass the value between the calling function and pivot_integer(). The function pivot_integer() must:

- Find the pivot integer x between 1 and n such that the sum of all integers from 1 to x, inclusively, is equal to the sum of all integers from x to n, inclusively.
- Display the pivot integer x if it exists.
- Display -1 if no such pivot integer exists.

CODE

```
#include <stdio.h>


void pivot_integer(int *n)
{
  int x, sum1, sum2;


  for (x = 1; x <= *n; x++)
  {
    sum1 = x * (x + 1) / 2;
    sum2 = (*n) * ((*n) + 1) / 2
        - (x - 1) * x / 2;


    if (sum1 == sum2)
    {
      printf("%d", x);
      return;
    }
```

```
    }

    printf("-1");
}

int main()
{
    int n;

    scanf("%d", &n);

    pivot_integer(&n);

    return 0;
}
```